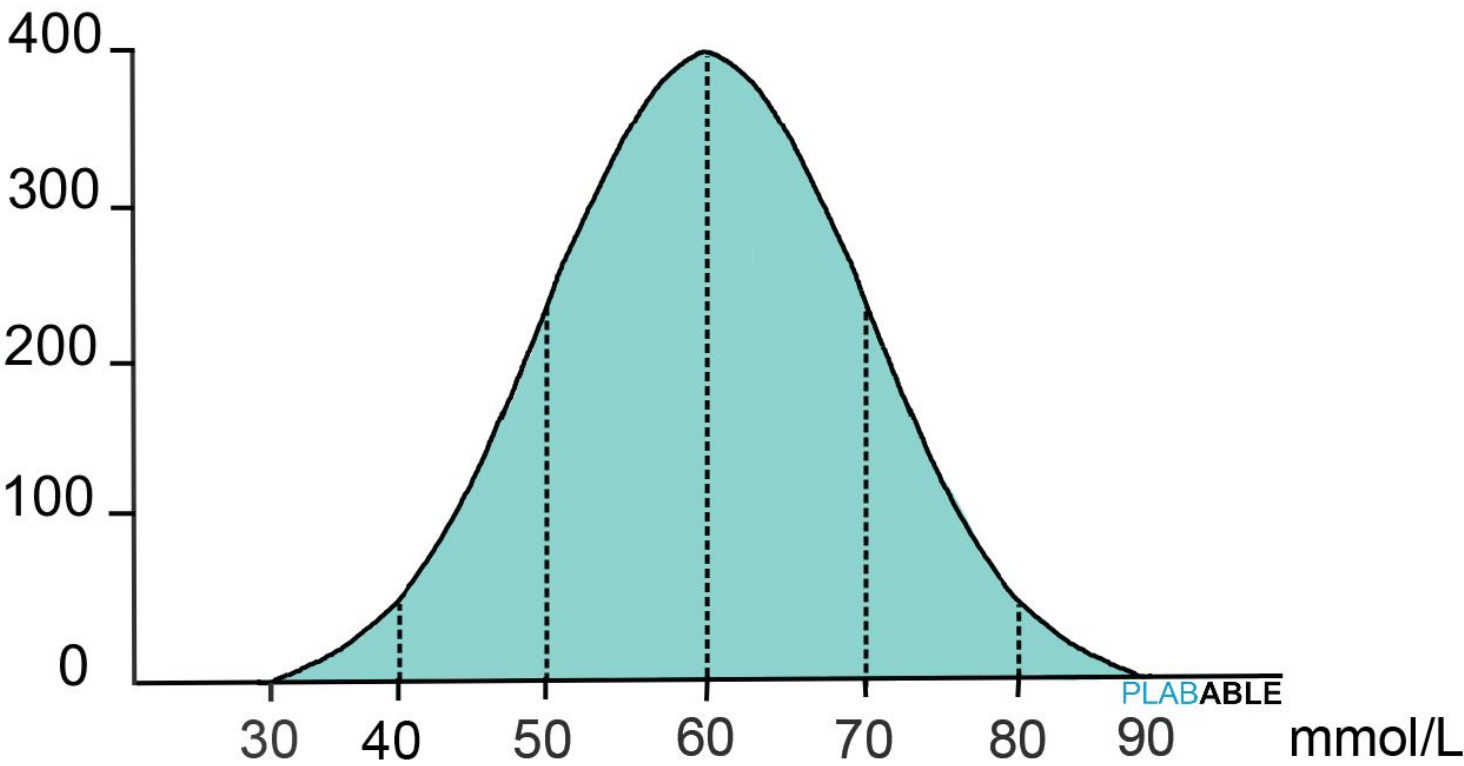


PLABABLE

GEMS 

VERSION 1.1

EPIDEMIOLOGY



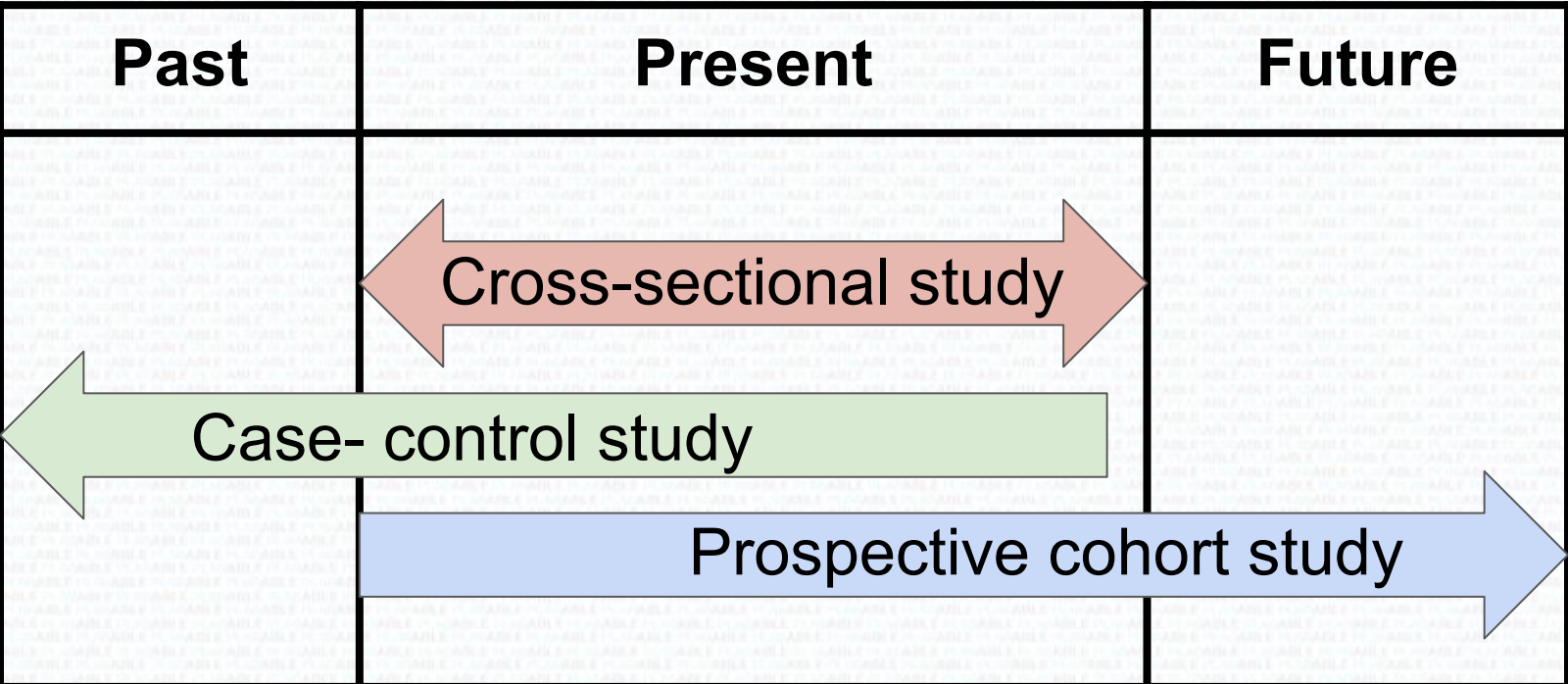
Observational Study

Cross sectional	Exposure & outcome taken at the same time in defined population	Compare diagnostic tests Check association between exercises & CHD
Case-control	To determine the cause of the disease Comparing cases with control population	Comparing lung cancer population with normal to find if smoking is a risk factor
Cohort	Groups, one with exposure and other without are observed	Comparing smokers and non-smokers over period of time to see how many develops lung cancer

Observational Study

Prospective: If groups followed up ahead

Retrospective: If observations recorded from history



Interventional Study

Randomised control trial

Participants are randomly assigned to two groups, one experimental and other control group

New drug trial

Other Types of Studies

Case study or Case series	Report of unique or rare case or cases
Systematic review	Review or compilation of all the articles on a particular topic
Meta-analysis	Mathematical analysis of results available from all published articles on particular topic

Type of Study

Brain trainer:

Two groups of women with osteoporosis are followed at certain intervals over a set period of time. One group is given a new test drug, the other group a placebo. What type of clinical study is described here?

→ **Randomised control trial**

Type of Study

Brain trainer:

A study was conducted to look for a connection between hard water and childhood eczema. During the study, group A received softened water and group B, normal tap water. The researchers did not know which group a patient was allocated into. What is the study design?

→ **Single-blinded randomised control trial**

Effectiveness of Screening

Brain trainer:

A new screening test for cervical cancer is being developed trialled on 10000 women. The screening test identified 1000 women with premalignant changes. Only 100 of these were confirmed (by biopsy) to have cervical cancer. How would you describe the effectiveness of this new test?

→ **High False Positives: Low Specificity**

Effectiveness of Screening

Brain trainer:

A new screening test for prostate cancer is being trialled on 1000 men. The new test reveals 43 men with a positive finding. On biopsy, only three of these findings are found to be negative, the remainder are positive. How would you describe the effectiveness of this new test?

→ **Low False Positives: High Specificity**

Effectiveness of Screening

Brain trainer:

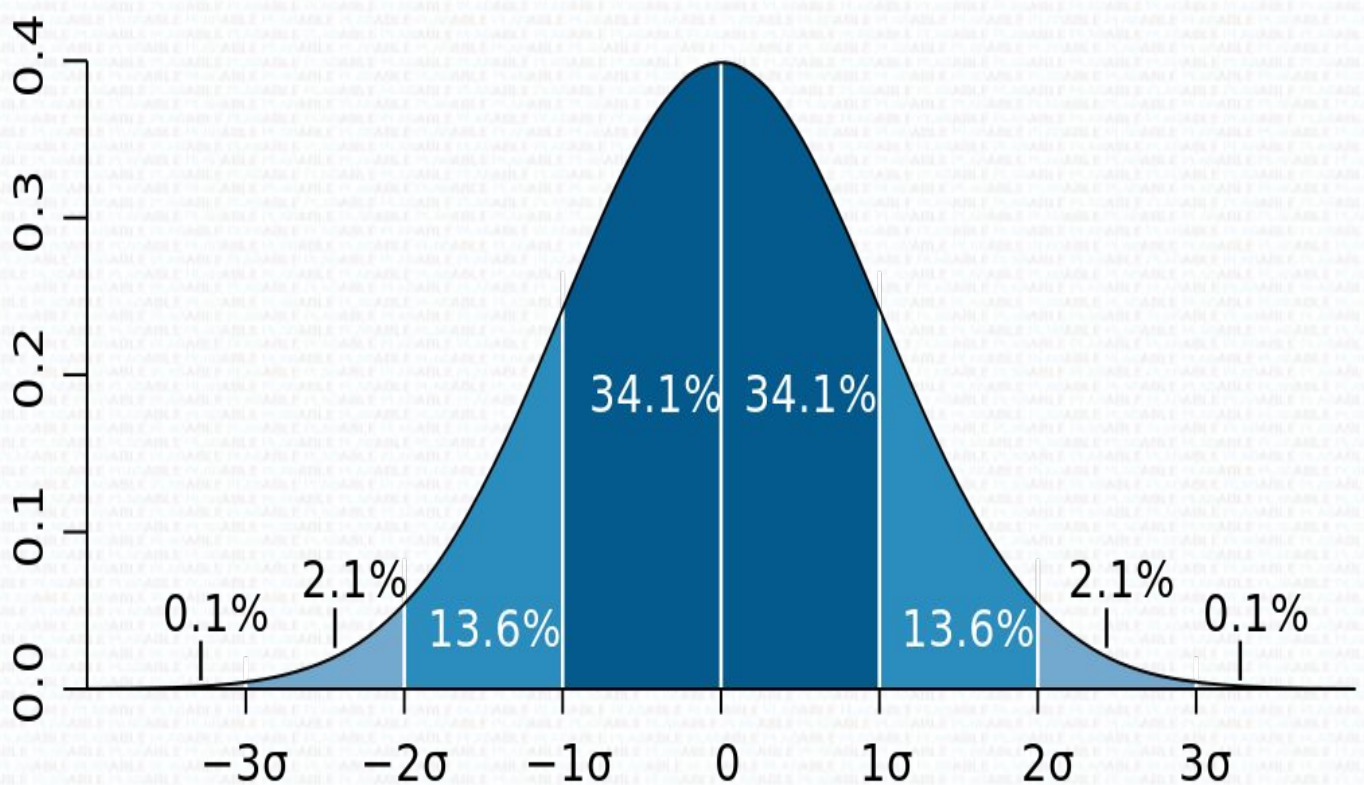
A new screening test for multiple sclerosis (Ms101) is being trialled on 5000 women. The new test is negative with 4900 women. The clinical picture and imaging reveals only 5 women with a negative screen are in fact positive. How would you describe the effectiveness of this new test?

→ **Low False Negatives: High Sensitivity**

Descriptive Measures

Standard Deviation (σ): A measure of dispersion of set of observations.

- It is a number used to tell how observations in a study are spreaded out when plotted on distribution curve



In normal distribution curve:

1 σ = 68.2%

2 σ = 95.4%

3 σ = 99.7% of observations

Descriptive Measures

Mean: Average value of all observations

Mode: Commonest occurring value among all observations

Median: A value which divides distribution in two equal parts

Example: 4, 2, 4, 3, 6, 1

- Mean = $(4+2+4+3+6+1)/6 = 20/6 = 3.33$
- Median = (1, 2, 4, 3, 6)
4 is in the middle so, median is 4
- Mode = 4 has occurred frequently, 4 is mode

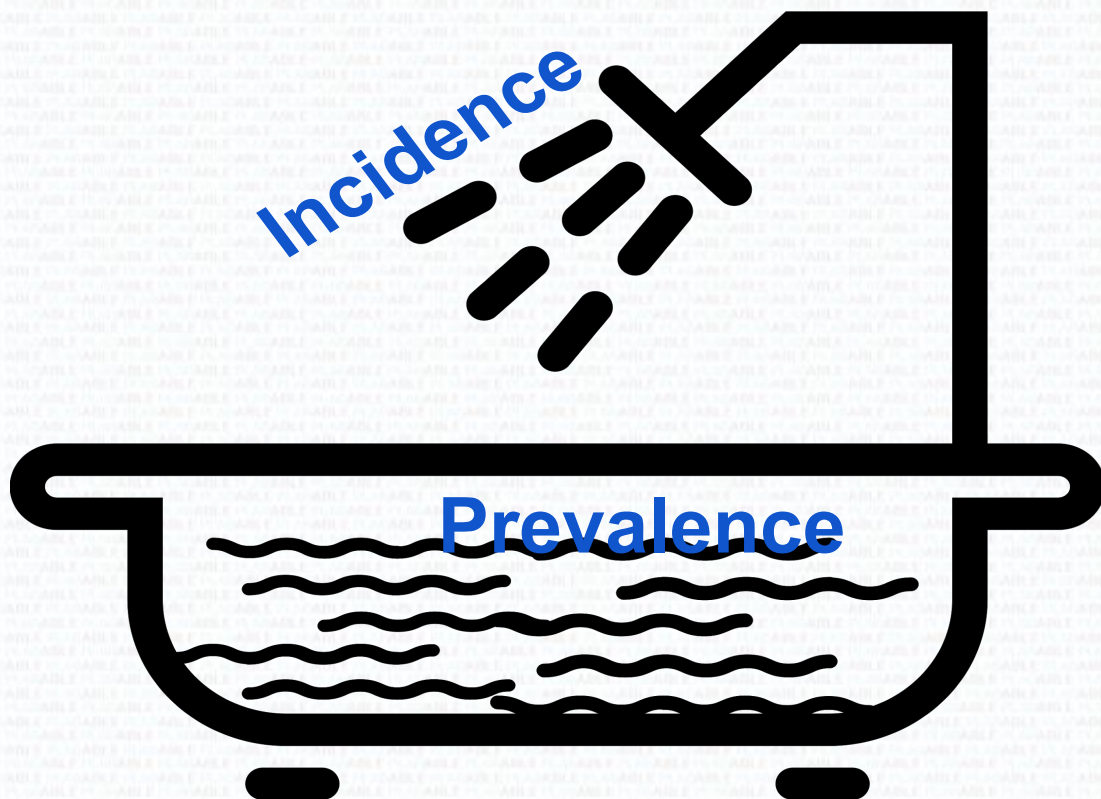
Measures of Morbidity

Incidence

Number of new cases occurring in specified time

Incident rate

New cases / total population at risk during that time



Prevalence

Number of affected persons present during specified time

Prevalence rate

Total cases / Total population during that time
(old+new)

Measures of Morbidity

Brain trainer:

It 52 cases of HIV were reported in the year 1995 in a population of 1000. The population was observed for two years and 28 new cases were reported during the observation period. What is the incidence and prevalence in the year 1997?

- Incidence rate is 28 per 1000 population
- Prevalence rate is $52+28 = 80$ per 1000 population

Measures of Mortality

Absolute Risk: Probability or chance of an event

	Disease	Non-disease
Exposure/ Treatment group	a	b
Non-exposure/ Control group	c	d

Absolute risk Treatment: ART
AR events in treatment group= $a/(a+b)$

Absolute risk Control: ARC
AR events in control group= $c/(c+d)$

Absolute Risk Reduction: Risk difference
= ARC - ART

= (mortality of control gr) - (mortality of case gr)

Convert percentage to decimal points

Measures of Mortality

Numbers needed to treat (NTT): number of cases needed to be treated to prevent additional bad outcome such as death

= $1 / \text{Absolute Risk Reduction}$

Brain trainer:

30 patients needed to use drug X to prevent lung cancer compared to a placebo drug in a study which continued for 15 years.

What is the NNT for the drug X?

→ 30

The lower the NNT, the better the drug

Relative Risk Reduction (RRR): how much the treatment reduced the risk of disease in control group

$(\text{ARC-ART}) / \text{ARC}$

Measures of Mortality

Relative Risk (risk Ratio): Probability of an event occurring in treatment group to the probability of an event occurring in control group

= **ART/ARC** = {a/(a+b)}/ {c/(c+d)}

	Disease	Non-disease
Exposure/ Treatment group	a	b
Non-exposure/ Control group	c	d

Odds Ratio: Odd of an event occurring in treatment group to odds of an event occurring in control group

$$\frac{a/b}{c/d} = ad/bc$$

Incidence

Brain trainer:

Each year 950 000 people develop myocardial infarction in a population of 250 000 000. Out of these, 215 000 die and 112 500 in the first hour. What is the incidence ?

→ 950 000 in 250 000 000

Absolute Risk

Brain trainer:

200 patients are taking a certain medication. 10 of them die whilst taking this drug. What is the absolute risk for those taking this drug?

→ $10 / 200 = 5\%$

Absolute Risk Reduction

Brain trainer:

- In a controlled group 10% of patients have a myocardial infarction. In the treatment group, 5% of patients undergo a myocardial infarction. What is the absolute risk reduction?
- **10% - 5% = 5%**

Relative Risk

Brain trainer:

A study into the development of COPD examines two groups: one group is comprised of non-smokers and the other group only smokers. The risk of developing COPD in the smoking group was 6% and in the non-smoking group 2%. What is the relative risk of developing COPD in the smoking group?

- $6 / 2 = 3$

This means that a smoking increases the risk of COPD by a factor of 3.

Relative Risk Reduction

Brain trainer:

	Number of patients who developed prostate cancer within the 5 year period	Number of patients who did NOT developed prostate cancer within the 5 year period
Prostagist	4	96
Placebo	10	90

What is the relative risk reduction ?

➔ 60%

ARC = Abs. risk in control group = 10 / (10+90)
= 0.1

ART = Abs. risk in treatment group = 4 / (4+96)
= 0.04

RRR = (0.1 - 0.04) / 0.1 = 0.6

This means there is a 60% reduction in the relative risk of prostate cancer in the treatment group compared with the control group.

False Negative

Brain trainer:

A study was carried out on 10 000 women. 1000 women had a positive smear. Of these, 100 developed cervical cancer. 10 women who had negative smear developed cancer. What is the term used to describe the 10 women with negative results who developed cervical cancer?

→ **False negative**

Prevalence

Brain trainer:

80 people are infected with a virus in a town. A short time later, 10 new cases are announced, 20 infected individuals die and 30 infected individuals are cured. The current population of this town is 1000.

What is the current prevalence of the virus in this town?

$$\rightarrow [(80 + 10 - 20 - 30) / 1000] \times 100 = 4\%$$

Study Analysis

Brain trainer:

A double-blind study was designed to test the efficacy of a drug. Halfway through the trial, 25% of participants in the placebo group dropped out of the study and only 3% from the other group. What is the best explanation for this occurrence?

→ Breakdown of the double-blind study

This most likely occurred as those taking placebo realised it was not an active substance and hence ceased to participate. The large discrepancy (25% vs 3%) supports this hypothesis.

Sensitivity & Specificity

Test Result \ Patients	Diseased	Healthy
Positive	True Positive TP	False Positive FP
Negative	False Negative FN	True Negative TN

Sensitivity = $TP / (TP + FN)$

Sick who test positive on test

Specificity = $TN / (TN + FP)$

Healthy who test negative on test

Positive Predictive Value = Proportion of total positive patients who are sick

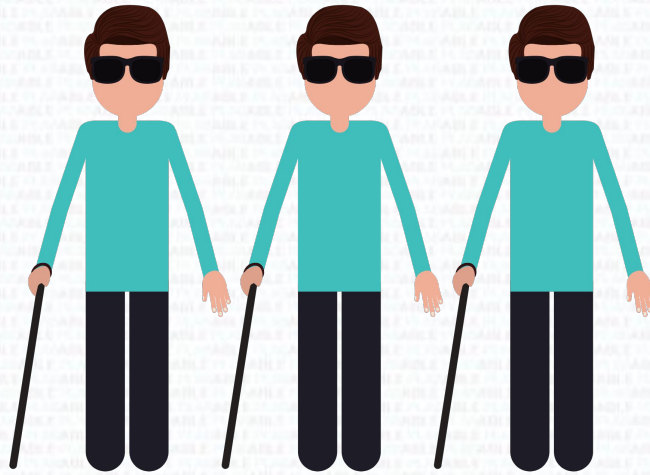
$PPV = TP / (TP + FP)$

Negative Predictive Value = Proportion of total negative patients who are healthy

$NPV = TN / (FN + TN)$

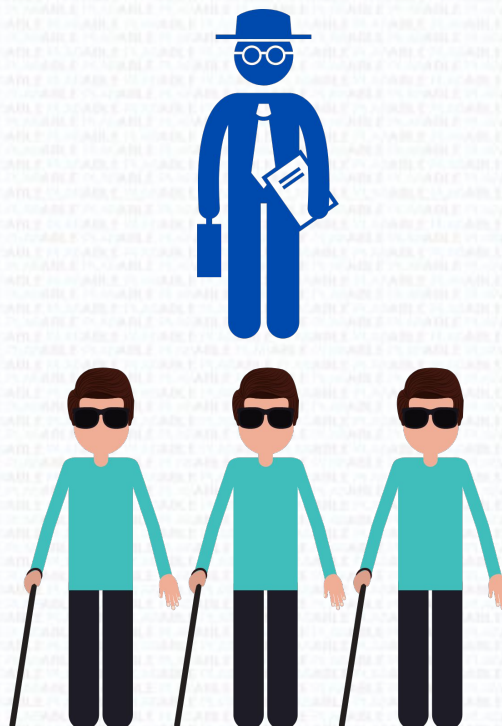
Blinding in Clinical Trials

Single blinded study: Participants alone



- Done to **reduce bias**
- Double blinding trials are superior as it reduces observer bias and outcome bias due to placebo effect

Double blinded study: Clinician + Participants



Epidemiological Measures

Index case: The first patient that indicates the existence of an outbreak

Primary case: The person who first brings a disease into a group of people

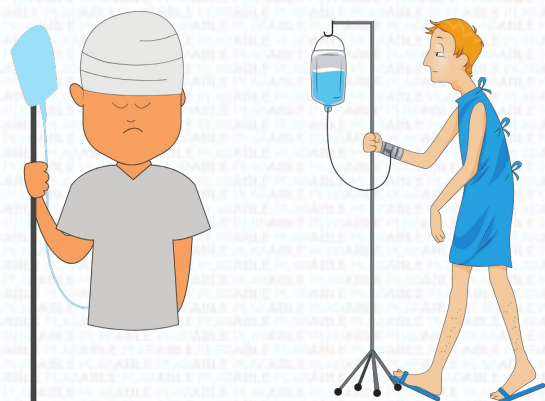
→ only apply to infectious diseases that spread from human to human

Secondary case: a person who gets a disease from exposure to a person with the disease, or primary case.

Primary
case



Secondary
case



Epidemiological Measures

Primary attack rate

Similar to attack rate

$$= (\text{Number of affected persons} / \text{Total population}) \times 100$$

Secondary Attack Rate

A number of exposed persons who develop the disease within incubation period following exposure to primary case

$$= (\text{Number of exposed persons who developed disease within incubation period} / \text{Total number of exposed contacts}) \times 100$$

Epidemiological Measures

Brain trainer:

A total of 50 people were on a board of an aircraft travelling to London. A week later, 5 passengers who were British nationals developed flu-like symptoms and were all tested positive for the new virus. This new virus was not previously seen in the UK. One incubation period later, 10 household members of positive passengers also tested positive for the same virus. If the total number of all the household members of the 5 passengers is 25 (inclusive of the 5 passengers), what is the secondary attack rate amongst household contacts?

$$\rightarrow [10/(25-5)] \times 100 = 50\%$$

Types of Prevention

Primary prevention: to prevent disease or injury before it ever occurs

Eg. Immunisation,
healthy habits like nutritious diet, regular exercises to prevent heart disease

Secondary prevention: to reduce the impact of a disease or injury that has already occurred

Eg. Screening tests to detect disease in its earliest stages, low-dose aspirin to prevent further heart attacks or strokes

Tertiary prevention: to soften the impact of an ongoing illness or injury that has lasting effects

Eg. Rehabilitation programs

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